Evidence of avoiding survivorship and look-ahead bias

Patrick Xiong (sx2434) - shared WRDS sample, three separately fitted targets

Scoped claim. Conditional on dated CRSP records/public-term dates representing historical information, this pipeline avoids constructing the past sample from end-period survivors and using future observations or unrealized labels in learning/decisions. This is finite executable evidence, not certification of all historical vendor release vintages or a prospectively untouched research design.

A. The historical universe is independent of eventual survival

Witness	Result
Official membership sample	705 historical PERMNOs; 20,675 cases; 41 decisions
Forecast retention	11,083 cases per target; no missing official member
Wrong end-period survivor screen	205 historical IDs / 3,351 rows would disappear; all retained
Actual historical terminal-event records	119 securities and 1,650 member-quarter rows retained
Erase future membership starts/end dates	Zero difference across all 41 dated universes
Named failed-bank outcomes	SIVB: 20 cases; SBNY: 5; FRC: 17; failure-quarter labels present

Universe(t) is constructed from effective membership spells at t, independently of quote presence and feature missingness. PERMNO separates securities whose tickers change or collide. Every member remains an inference case even when its feature vector or eventual outcome is missing.

B. Missing outcomes cannot silently remove eligible failures

Supervised eligibility uses an executable start quote and already-public rights terms. Every eligible completed-horizon outcome must exist; otherwise the trainer stops. Eligibility never asks whether the stock survived, remained in the final index or eventually had a finite answer.

Case outside target domain	Decision-time information
ABMD: December 1,2022	CVR fact available November 4,2022
WBA: June 2,2025	CVR fact available March 11,2025
INFO: March 1,2022	No starting quote after February 28 terminal record
503 final-quarter outcomes	Horizon not complete at September 30,2025

All 20,169 eligible completed-horizon stock labels are present. The other 506 cases remain predictions. Deliberately setting an eligible SIVB failure label to missing causes an error. Unknown CVR complete consideration is not replaced by zero or an invented payoff.

C. Accounting recognizes loss only when the source observation exists

CIZ terminal daily returns are used once. Deferred OTC quotes or payments move recognition to the later observed market date. SIVB/SBNY reconstruction is limited to documented PRCF subsequent-price events, zero cash component and a checked pre-event price. Frozen held dollars remain until execution; unresolved held proceeds/gaps are fatal. A separate dollar ledger recomputes losses, cash and fees.

D. Actual fit records and counterfactual information tests

For decision t, feature observations satisfy s < t. At training cutoff c, eligible labels and every source observation required to establish them must be available before c. Validation labels must be observed strictly before t. Event-recognition dates and verified quote-resumption dates are included in the source clock.

Check	Executed result
Same data across SPY/absolute/QQQ targets	Identical case keys, six feature arrays, original stock returns and train/validation partitions
Delete future raw rows before feature construction	20,675 cases; maximum feature difference 0
Actual fit/score/predict row-key reconciliation	858 records per target,2,574 across three targets; zero temporal breaches
Saved models and train-only imputation	286 estimators per target; 858 separately fitted estimators total
Poison unavailable covariates/labels	No change in admitted training values or targets
Poison inference targets/future-status fields	Input matrices identical; predictions differ by less than1e-12
Leaky global-median negative control	Detected across all 286 estimators per target
Independently reconstruct QQQ-relative targets	20,169 labels; maximum error 1.78e-15
Independently reconstruct portfolio NAV	21 QQQ experiment portfolios x 1,361 dates; max difference 3.24e-14

Forecast	Training cutoff	Latest train label	Latest validation label
2020-06-01	2019-06-03	2019-03-01	2020-03-02
2025-09-02	2024-09-03	2024-06-03	2025-06-02

Reproduce and inspect

Run code/reproduce_all.py with the four licensed WRDS inputs. Bias witness CSVs and summaries are in evidence/bias_proof and experiments/{qqq, absolute}/evidence/bias_proof. Raw/prepared data and estimator weights stay outside the submission archive. The review notebook reads compact result snapshots without requiring a WRDS account.

Limits. Original-vintage vendor revisions and individual membership-announcement timestamps are not reconstructed. The dated rights rule is developed and nonexhaustive; unrestricted all-stock representativeness is not proved. Strategy/target adoption used this history, so statistical alpha, untouched-holdout status and absence of research-selection bias are not claimed. A read-only Claude Opus 5.5 audit (October 12, 2026) rebuilt the universe, labels, features, fit partitions and NAVs from the frozen inputs and re-ran the packaged non-training scripts. It did not refit models or certify CRSP release vintages.